12 A student models a playground swing as a simple harmonic oscillator moving along a straight horizontal path directly above a scale.

The extreme positions reached by the swing correspond to the 0 m mark and the 2.00 m mark on the scale.

If the period of oscillation of the swing is 2.4 s, how long does it take for the swing to travel directly from the 0.75 m mark to the 1.50 m mark?

A 0.30 s **B** 0.40 s **C** 0.50 s **D** 0.90 s